

Class 14 Intro to Groups

Goals:

1. Understand what is meant by algebraic structure.
2. Know definition of a group.
3. Become familiar with examples of groups.

Definition: An algebraic structure is a non empty set with at least one operation satisfying a specified list of properties.

Definition: A group is a nonempty set, G , with one binary operation, $\circ$, satisfying 4 properties.

1. Closure: $\forall x, y \in G, x \circ y \text{ is in } G$

2. Associativity: $\forall x, y, z \in G \quad (x \circ y) \circ z = x \circ (y \circ z)$

3. Identity: $\exists$ an element e in G such that $x \circ e = x = e \circ x \quad \forall x \in G.$

4. Inverse: $\forall x \in G, \exists y \in G$ such that $x \circ y = e = y \circ x.$
write $y = x^{-1}$

Examples: $(\mathbb{Z}, +)$ integers under $+$

$(\mathbb{Z}_n, \oplus_n)$ integers mod n under addition mod n

$(\mathcal{U}(n), \odot_n)$ Positive integers less than n and relatively prime to n under mult mod n .

$(\mathbb{Q}^*, \cdot)$ nonzero rational numbers under multiplication.

Why do the following sets fail to form a group under the given operation?

Integers under subtraction:

not associative

$\mathbb{Z}_n$ under regular addition:

not closed

Rational numbers under multiplication:

No inverse for 0

Irrational numbers under addition:

Not closed, no identity element

Three properties every group has:

Unique Identity Theorem: The identity element in any group is unique.

Unique Inverse Theorem: In any group, G , the inverse of any element, a , will be unique.

Left Cancellation Theorem: For any elements a , b , and c in a group, G , if $a \circ b = a \circ c$ then $b = c$.

Proofs

1. Every group has a unique identity element.

Suppose for contradiction that G had two distinct identity elements e and f .

Then $e = e \circ f$ since f is an identity
and $f = e \circ f$ since e is an identity
so $e = f$ contradicting that they
were distinct. Thus G has a unique
identity element. $\square$

2. If a is any element of G then a has a unique inverse element.

Let a be any element of G and suppose for
contradiction that there are two distinct elements
 x and y in G such that $a \circ x = e = x \circ a$ and
 $a \circ y = e = y \circ a$

Then $x = x \circ e = x \circ (a \circ y) = (x \circ a) \circ y = e \circ y = y$.

This contradicts that x and y are distinct.

Thus every element of G has a unique
inverse element. $\square$

Generally denote the unique inverse of a by a^{-1} . If the operation is addition we denote the inverse of a by $-a$.

3. If a , b , and c are any elements of a group and $a \circ b = a \circ c$, then $b = c$.

Let a, b, c be in group G and let $a \circ b = a \circ c$.

Multiplying both sides by a^{-1} we get

$$a^{-1} \circ (a \circ b) = a^{-1} \circ (a \circ c).$$

Reassociate to get $(a^{-1} \circ a) \circ b = (a^{-1} \circ a) \circ c$.

which leads to $e \circ b = e \circ c$.

So $b = c$

